

Data Scientist - Economist

Summary

Data scientist at the French Senate (previously at the General Inspectorate of Finance and the French Treasury). Specialised in fraud detection, public policy evaluation and the analysis of large-scale datasets (SNDS, tax data, national accounts). Over 10 years of experience in quantitative studies, impact assessments and the development of decision-support tools (R, Python, Spark, Shiny, cloud).

Professional experience

- 2025 – **Data Scientist**, French Senate, Paris
- **Evaluation work**: quantitative support for the evaluation and oversight of government action (data analysis, impact assessments, drafting of briefing notes).
 - **Legislative work**: quantitative analyses, simulations and summaries on economic, social and budgetary topics.
- 2021 – 2025 **Data Scientist**, General Inspectorate of Finance (IGF), Paris
- **Fraud detection** (energy transition, international trade): developed a rule engine and a reference database containing several million prices from open sources (*web scraping*); built a *no-code* application enabling business teams to reproduce the analyses autonomously. *R, Shiny*
 - **Evaluation of the national healthcare strategy**: used the SNDS and DREES databases to analyse several billion healthcare claims and simulate reforms of the long-term condition scheme (ALD). *Spark, R, microsimulation, public finances*
 - **Decision-support tools**: developed two Shiny applications (HR management; mission tracking and GDPR compliance) and an interactive dashboard of central government revenues (Eurostat data) used by the Budget Directorate as part of the 2024 spending reviews. *R, Shiny, Python*
 - **Public policy evaluation**: several economic studies (tobacconists, nursing homes, financial sector; see publications).
- 2018 – 2021 **Deputy Head of Unit FIPU2**, *FIPU Subdirectorate*, French Treasury, Paris
- Corporate tax analysis and costing; major contribution to assessments and negotiations on the introduction of a global minimum tax promoted by the OECD ([BEPS project](#)).
 - Modernised and automated the production of general government accounts.
- 2015 – 2018 **Head of Investment and Tax**, *Quarterly Accounts Department*, Insee, Paris
- Oversaw the quarterly estimation of investment and taxes using econometric models and short-term indicators.
 - Produced tax revenue forecasts for Insee's economic outlook publication.

Technical skills

Programming & data	R (Shiny, Quarto, data.table), Python (pandas, scikit-learn), SQL, Spark, Git
Web	HTML, CSS, JavaScript, Bootstrap 5
Statistics / ML	Econometrics, predictive modelling, time series, microsimulation
Economics	Public finances, national accounting, macroeconomics

Infrastructure / Application containerisation (Docker), deployment and CI/CD pipelines (Git-Lab), security and GDPR, large-scale data processing (Spark, distributed databases)

Education

- 2019 **Master's degree (M2) – Public Policy Evaluation and Decision-Making**, ENSAI, Rennes (35)
French National School for Statistics and Information Analysis
- 2013 – 2015 **Insee statistician officer programme**, ENSAI, Rennes (35)
- 2011 – 2013 **MPSI/MP preparatory classes (maths and physics)**, *Lycée Alphonse Daudet*, Nîmes (30)

Publications

- 2025 [Additional construction costs linked to regulatory standards](#)
- 2025 [Enforcing sentences: 20 proposals to improve enforcement and give meaning to criminal sanctions](#)
- 2025 [Transparency and evaluation of state aid to businesses: a democratic demand and a key to economic efficiency](#)
- 2025 [Changes in the financing of efforts to combat violence against women](#)
- 2025 [25 proposals to give municipal police the tools to tackle everyday insecurity](#)
- 2024 [Spending review on long-term conditions: towards a more efficient and equitable scheme](#)
- 2023 [Cross-cutting methodological mission on the State's fields of intervention and public policy instruments](#)
- 2022 [Mission on the management of nursing homes \(EHPAD\) in the Orpea group](#)
- 2021 [Evaluation committee for the France Relance recovery plan](#)
- 2020 [New rules for the international taxation of corporations: assessment of the economic effects for France](#)

Full list available at www.philippefontaine.eu

Interests

Cybersecurity (TryHackMe, HackTheBox), web application development and open-source projects

Defence, geopolitical issues and social psychology

Electronic music production